

Training-Free Token Reduction on MotionBench

Benchmark Results for Eleven Methods across Two Backbones

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Abstract

This report presents benchmark results for eleven training-free video-LLM token-reduction methods on MotionBench (8,052 questions, 4,018 scoreable), evaluated on two backbones at a standardized 32 frames and 15% nominal token retention. Headline numbers: the Qwen3-VL-8B baseline reaches 62.52% against LLaVA-OV-7B’s 52.66%, a 9.86-point gap that holds across all six question categories. On LLaVA-OV no method differs from its backbone by more than the ± 1.54 -point resolution floor except two that collapse to 36.78% and 38.20%. On Qwen3-VL, at identical settings, all ten portable methods lose accuracy, ranging from -0.35 to -6.55 , and eight of those losses are statistically significant. Results are reported overall and by category; methodology, verification, and limitations are in the appendices.

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1 Setup

Benchmark. MotionBench: 8,052 four-option multiple-choice questions over video clips. 4,034 carry an NA ground truth and are excluded, leaving **4,018 scoreable items** — the denominator

for every accuracy figure below. Six question categories, of unequal size (Table 2). Scoring letter-matches A–D against ground truth; generation is greedy.

Backbones. **LLaVA-OV-7B** (SigLIP-so400m vision tower, Qwen2-7B LLM, mid-2024) and **Qwen3-VL-8B** (Qwen3VLForConditionalGeneration, a native architecture rather than a LLaVA derivative, late-2025).¹

Configuration. All methods run at 32 frames. Methods exposing a single retention parameter are set to 15%; those whose budget is fixed internally, or that select frames rather than tokens, keep published defaults. Comparison is therefore at a common *nominal* setting, not at equal compute — measured budgets range from 7.5% to 50% (Appendix A).

Reading the numbers. At $n = 4,018$ the binomial 95% confidence half-width is ± 1.54 points; smaller differences are not resolvable and are not ranked. * marks a difference significant by McNemar’s test on paired predictions ($\chi^2 \geq 3.84$, $p < 0.05$); paired counts are in Appendix B. Every figure here was recomputed from per-sample prediction records (Appendix A).

2 Results

2.1 Summary

Table 1: All methods on both backbones, sorted by Qwen3-VL result. * = significant vs. that backbone’s baseline. Dashes mark cells not independently recomputed for this report (see Appendix A) or not applicable.

Method	LLaVA-OV-7B		Qwen3-VL-8B	
	Overall	Δ	Overall	Δ
<i>Baseline</i>	<i>52.66</i>	—	<i>62.52</i>	—
PruneVID [†]	38.20	−14.46*	62.17	−0.35
DyCoke	53.36	+0.70	61.85	−0.67
HoliTom	53.14	+0.47	60.33	−2.19*
MDP3	—	—	59.66	−2.86*
FastV [†]	36.78	−15.88*	59.01	−3.51*
VisionZip (ctx.) [‡]	—	—	58.81	−3.71*
STTM	—	—	57.07	−5.45*
FlashVID	53.31	+0.65	56.65	−5.87*
VideoITG	—	—	56.35	−6.17*
AIM	—	—	55.97	−6.55*

[†] These two LLaVA-OV results are under review: both collapse with matched signatures and share one code path, and a control experiment separating method from harness has not yet reported (§2.7, Appendix D). [‡] Contextual half only — Qwen3-VL’s vision tower has no CLS token, so VisionZip’s dominant half is not implementable. Not the published method. VisionZip’s complete form runs on LLaVA-1.5-7B (Table 7). DyTo appears in Table 7: it is bound to a Vicuna backbone and runs on neither of these two.

¹Project records refer to a `qwen.1.5` template on LLaVA-OV. That is a prompt template, not a different LLM: `qwen.1.5` and `qwen.2` render byte-identical prompts here and the model is Qwen2-7B either way.

Table 1 contains the report’s main observation. Reading across rows rather than down columns, the ordering does not survive the change of backbone: PruneVID is the worst result on LLaVA-OV and the best on Qwen3-VL; FlashVID is indistinguishable from the backbone on LLaVA-OV and loses 5.87 points on Qwen3-VL. Every Qwen3-VL delta is negative; three of five LLaVA-OV deltas are positive but none resolvably so.

2.2 Backbones

Table 2: Baseline accuracy overall and by category (% correct within category). AO = Action Order, CM = Camera Motion, LM = Location-related Motion, MR = Motion Recognition, MO = Motion-related Objects, RC = Repetition Count. Chance is 25%.

Backbone	Overall	AO	CM	LM	MR	MO	RC
LLaVA-OV-7B	52.66	40.5	45.2	55.5	57.0	71.2	23.8
Qwen3-VL-8B	62.52	46.1	63.1	65.0	67.3	79.0	33.8
Δ	+9.86	+5.6	+17.9	+9.5	+10.3	+7.8	+10.0
n	4,018	519	385	546	1,478	690	400

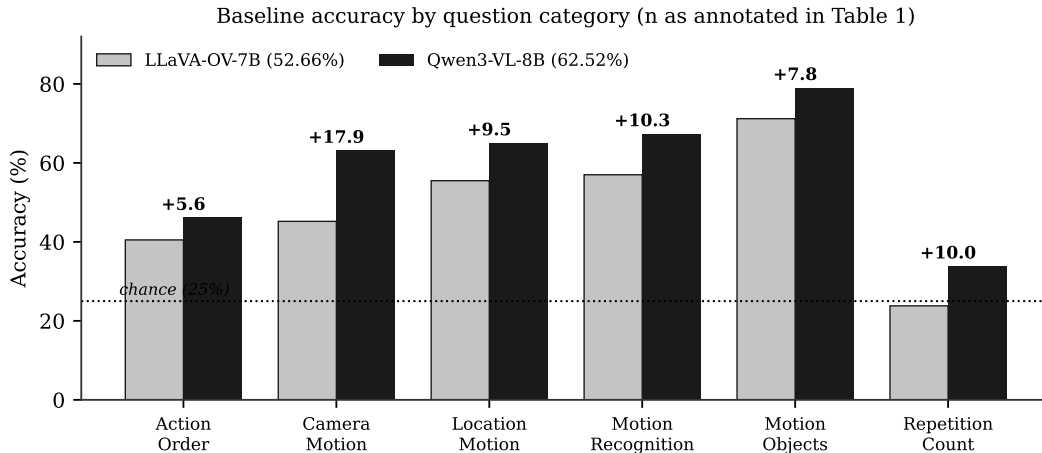


Figure 1: Baseline accuracy by category. Qwen3-VL-8B leads in all six, most sharply on Camera Motion (+17.9).

The 9.86-point backbone gap is larger than any method effect measured anywhere in this report, and it holds in every category.

Two properties of Table 2 govern how later tables should be read. Both backbones are strongest on Motion-related Objects (71.2%, 79.0%), the category most answerable from object appearance without integrating motion. And **Repetition Count is at or near chance for both** (23.8% and 33.8% against 25%): LLaVA-OV is not distinguishable from guessing there, so movement in that column is not evidence about a method and is not interpreted below.

2.3 LLaVA-OV-7B

Table 3: LLaVA-OV-7B, 32 frames, 15% nominal retention. Categories as in Table 2. VisionZip runs on a different backbone (Table 7); MDP3, AIM, VideoITG and STTM are not included here (Appendix A).

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
DyCoke	53.36	+0.70	38.9	48.8	55.7	58.2	70.9	25.2
FlashVID	53.31	+0.65	39.9	45.7	53.8	58.0	71.7	28.2
HoliTom	53.14	+0.47	40.8	49.6	52.6	57.0	71.4	27.3
<i>Baseline</i>	<i>52.66</i>	—	<i>40.5</i>	<i>45.2</i>	<i>55.5</i>	<i>57.0</i>	<i>71.2</i>	<i>23.8</i>
PruneVID-OV* [†]	38.20	−14.46	32.0	34.8	35.5	38.4	52.9	27.3
FastV* [†]	36.78	−15.88	32.8	31.9	34.1	35.7	53.8	25.2

DyCoke, FlashVID and HoliTom land within 0.70 points of the backbone, inside the ± 1.54 -point floor, none significant (largest $\chi^2 = 2.34$ against a 3.84 threshold). They are statistically indistinguishable from the backbone and from each other; the ordering among them is not meaningful. Their category profiles track the backbone closely, with the largest deviations falling in the two smallest categories.

FastV and PruneVID-OV lose more than 14 points each, significantly, and degrade in every category carrying above-chance backbone signal (§2.7).

2.4 Qwen3-VL-8B

Table 4: Qwen3-VL-8B, 32 frames, 15% nominal retention. All ten portable methods. VisionZip is the contextual half only (see Table 1 notes).

Method	Overall	Δ	AO	CM	LM	MR	MO	RC
<i>Baseline</i>	<i>62.52</i>	—	<i>46.1</i>	<i>63.1</i>	<i>65.0</i>	<i>67.3</i>	<i>79.0</i>	<i>33.8</i>
PruneVID	62.17	−0.35	46.1	62.9	64.1	67.1	79.1	32.5
DyCoke	61.85	−0.67	45.1	62.6	63.9	66.4	78.4	34.5
HoliTom*	60.33	−2.19	44.7	61.3	61.0	66.4	76.2	29.0
MDP3*	59.66	−2.86	44.9	64.7	62.8	64.0	77.4	23.0
FastV*	59.01	−3.51	46.2	61.0	61.9	63.5	72.2	30.5
VisionZip (ctx.)*	58.81	−3.71	43.2	57.4	61.9	64.2	74.3	29.5
STTM*	57.07	−5.45	44.3	60.0	61.0	60.6	73.8	23.8
FlashVID*	56.65	−5.87	43.4	57.9	57.1	61.2	73.0	27.0
VideoITG*	56.35	−6.17	45.3	53.5	59.3	60.2	75.4	22.2
AIM*	55.97	−6.55	45.9	56.6	59.5	58.3	72.2	27.0

Every method loses accuracy on Qwen3-VL and eight of ten lose significantly. The two exceptions, PruneVID (−0.35) and DyCoke (−0.67), are the two that retain roughly half their tokens against 7.5–15% for the rest (Appendix A).

Loss does not track budget monotonically. HoliTom operates at the smallest measured budget in the set (7.5%) and loses 2.19 points, while AIM and FlashVID at 15% lose 6.55 and 5.87. The methods that hold up are those that merge non-retained tokens into survivors rather than discarding them outright.

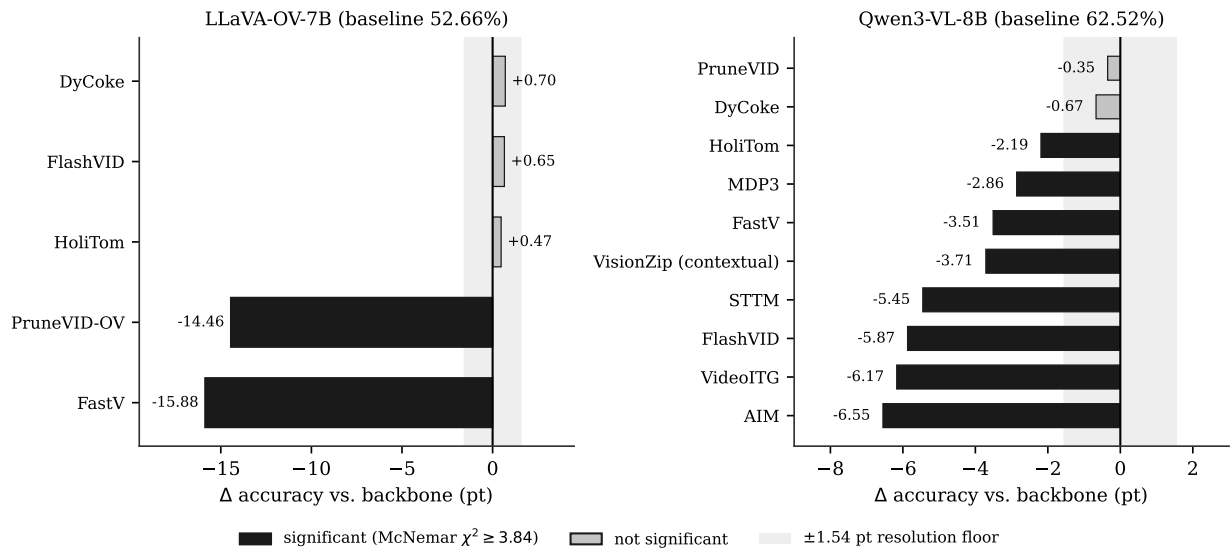


Figure 2: Change vs. backbone on each model. Shaded band is the ± 1.54 -point resolution floor. On LLaVA-OV every non-collapsed method falls inside it; on Qwen3-VL eight of ten fall outside it and all ten are negative.

2.5 Per-category change

Losses are not spread evenly across question types. Expressing retained above-chance signal as $(\text{method} - 25) / (\text{backbone} - 25)$, FastV on LLaVA-OV keeps **62%** of the backbone’s above-chance signal on Motion-related Objects but only **30%** on Location-related Motion, 33% on Motion Recognition and 34% on Camera Motion. PruneVID-OV shows the same ordering (60% on Motion-related Objects, 34% on Location-related Motion). Motion-related Objects is the category most answerable from appearance in isolated frames; categories requiring integration across frames retain roughly half as much.

The pattern is milder but present on Qwen3-VL (Figure 3, right), where the largest losses fall on Motion Recognition (-9.0 , AIM) and Camera Motion (-9.6 , VideoITG), while Action Order moves within ± 3 points for every method.

Repetition Count is the control column. LLaVA-OV scores 23.8% there against 25% chance, so there is no signal to preserve; the apparent gains under FastV (+1.5) and PruneVID-OV (+3.5) accompany 15-point overall collapses and reflect answer redistribution (§2.7), not recovered capability.

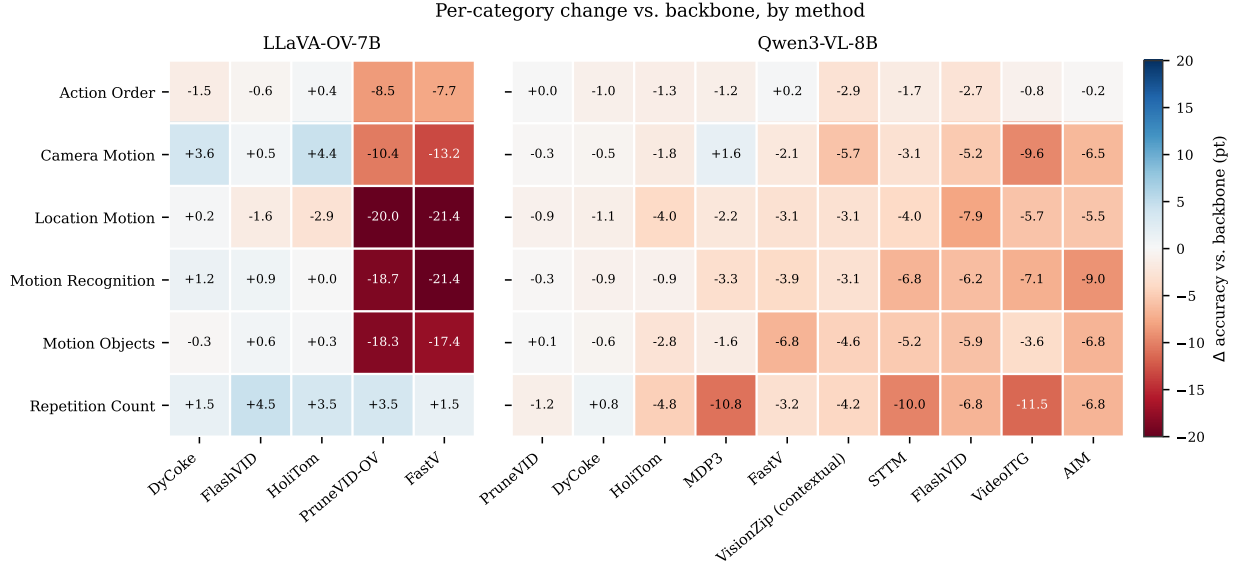


Figure 3: Per-category change vs. backbone, columns ordered by overall delta. Repetition Count movement is not interpretable — both backbones are at or near chance there (Table 2).

2.6 Retention sweep

Table 5: Accuracy (and Δ vs. backbone) across nominal retention, 32 frames and all else fixed. Span is movement from 10% to 25%. Four of eleven methods expose a comparable retention parameter.

Method	$r = 0.10$	$r = 0.15$	$r = 0.25$	Span
<i>LLaVA-OV-7B (baseline 52.66%)</i>				
FastV	36.06 (−16.60)*	36.78 (−15.88)*	36.73 (−15.93)*	+0.67
FlashVID	52.51 (−0.15)	53.31 (+0.65)	53.36 (+0.70)	+0.85
HoliTom	52.76 (+0.10)	53.14 (+0.47)	53.41 (+0.75)	+0.65
PruneVID	38.10 (−14.56)*	38.20 (−14.46)*	38.38 (−14.29)*	+0.28
<i>Qwen3-VL-8B (baseline 62.52%)</i>				
FastV	56.92 (−5.60)*	59.01 (−3.51)*	60.60 (−1.92)*	+3.68
FlashVID	54.73 (−7.79)*	56.65 (−5.87)*	59.36 (−3.16)*	+4.63
HoliTom	60.05 (−2.46)*	60.33 (−2.19)*	60.43 (−2.09)*	+0.38
PruneVID	60.23 (−2.29)*	62.17 (−0.35)	61.40 (−1.12)*	+1.17

Tripling the budget from 10% to 25% moves no LLaVA-OV method more than 0.85 points. On Qwen3-VL the same change moves FlashVID 4.63 points and FastV 3.68, with FastV’s significance decaying monotonically ($\chi^2 = 89.8 \rightarrow 44.6 \rightarrow 18.2$) as it converges toward the baseline — roughly a five-fold difference in retention sensitivity between the two backbones.

On LLaVA-OV the four methods sit in two flat regimes: FlashVID and HoliTom at the baseline at every setting, FastV and PruneVID roughly 15 points below it at every setting. What costs accuracy there is that those two prune at all, not how hard.

One cell is flagged rather than smoothed: PruneVID on Qwen3-VL is non-monotonic (60.23 $\rightarrow$ 62.17 $\rightarrow$ 61.40), its midpoint both its best result and its only non-significant one. At ± 1.54 -point resolution this may be noise; it is not reported as a trend.

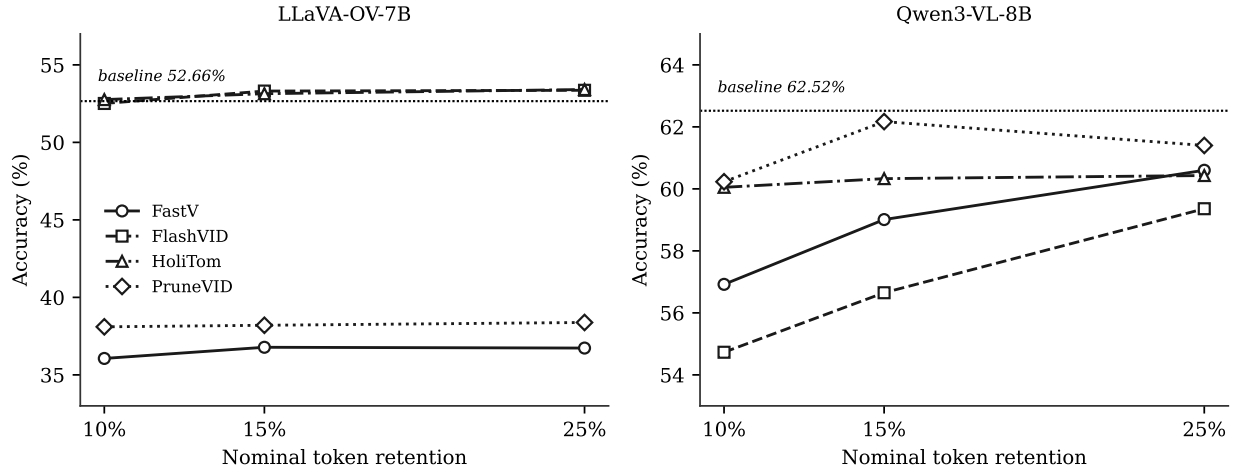


Figure 4: Accuracy against nominal token retention. On LLaVA-OV the four methods occupy two flat regimes the budget does not move; on Qwen3-VL the same methods respond strongly to the same parameter.

2.7 Answer distribution on LLaVA-OV

Table 6: Distribution of predicted answer letters on LLaVA-OV-7B, % of 4,018 scoreable items.

Run	A	B	C	D
<i>Ground truth</i>	25.4	25.6	25.5	23.6
Baseline	27.1	26.6	26.9	19.3
DyCoke	27.4	26.1	27.0	19.3
FlashVID	27.7	25.6	25.7	20.8
HoliTom	26.1	24.1	27.2	22.5
PruneVID-OV	12.2	26.7	15.0	46.0
FastV	14.9	23.0	15.2	46.7

The two collapsed methods place 46–47% of their answers on option D against a 23.6% ground-truth rate, while the backbone and the three merge-based methods stay near uniform. This is the behaviour of a model answering from the prompt without usable visual evidence, and it explains the apparent Repetition Count gains in §2.5: a near-uniform category rewards a degenerate constant answer at roughly chance.

The signature is retention-invariant. Across $r \in \{0.10, 0.15, 0.25\}$ FastV’s D-rate is 47.0%, 46.8%, 47.1% — flat to a tenth of a point while accuracy moves 0.67 points. Restoring tokens does not restore visual grounding on this backbone, though it does for the same algorithm on Qwen3-VL, where the answer distribution tracks the backbone within 1.5 points at every budget.

Caveat on these two rows. FastV and PruneVID-OV are different algorithms at different layers with different selection rules, yet fail with signatures matching to about one point on accuracy, D-rate, broken/fixed ratio and flatness across retention. Both LLaVA-OV ports drive reduction through the same code path. A control experiment separating algorithm from harness is queued and has not reported; until it does, these two rows are reported as measurements of this harness and their attribution to the published methods is open (Appendix D).

2.8 Methods on non-standard backbones

Table 7: Runs on backbones other than the two standardized ones. Not comparable to Tables 3–4; no Δ is given because no matched baseline exists for either row.

Method	Backbone	Overall	AO	CM	LM	MR	MO	RC
VisionZip	LLaVA-1.5-7B	40.09	33.7	33.0	37.2	41.1	57.4	25.8
DyTo (recon. TW-FINCH)	Vicuna-7B	42.06	32.4	33.0	42.1	43.0	61.7	26.0

VisionZip patches `CLIPVisionTower`, which LLaVA-OV does not have, so its complete form runs only on LLaVA-1.5-7B — and at 8 frames rather than 32. The 40.09% differs from Table 3 in both backbone and frame count and is never placed in a LLaVA-OV comparison.

DyTo carries two caveats. It is not reproducible from its published artifacts (the released code calls an argument absent from its own pinned dependency, and a clustering helper has no return statement); the missing return was recovered by transcription from a character-identical sibling function, but the temporal weighting had to be reimplemented, so the result is reported only as “DyTo (reconstructed TW-FINCH)”. And it is the one number in this report with execution evidence but **no divergence check**, because no baseline was ever run on its Vicuna backbone — it is an absolute score, not a measured method effect.

A token-matched control is documented in the project records (cited, not recomputed here): against an uncompressed 6-frame baseline at 3,456 visual tokens scoring 43.35%, DyTo at $\sim$ 3,680 tokens scores 42.06%, a significant -1.29 . A frame-matched control is impossible — Vicuna’s 4,096-token context cannot hold 32 uncompressed frames — which is the constraint DyTo exists to address.

A Methodology and verification

The verification gate

A run that completes, reports a plausible accuracy, and did nothing is indistinguishable from a successful one by accuracy alone. Every number in this report therefore required two independent signals.

1. Execution evidence. The method’s instrumentation reported a token count consistent with its configured budget (Appendix A).

2. Prediction divergence from the unmodified backbone. Greedy decoding makes the pipeline a deterministic function of (weights, frames, prompt): frame indices come from a fixed linear spacing, the vision tower is frozen, token choice is arg max. Re-running an unchanged configuration reproduces all 8,052 predictions exactly — confirmed on three independent run pairs, each diverging on 0 of 8,052. Against that zero baseline, a method that discards or merges most of the visual input and produces zero differing predictions cannot have executed.

The second signal is not redundant. Four ports passed every other check and failed it: an early FastV integration whose pruning call was a stub; a PruneVID port to LLaVA-OV whose discard mask was dropped before reaching the model; and FlashVID and AIM on Qwen3-VL, which logged a correct budget (`keep=1750 (15.0%)`) while emitting byte-identical output, because `attention_mask` is `None` under this model’s default attention implementation and the mask-dependent path never ran. A fifth, STTM on Qwen3-VL, failed because that model interleaves timestamp text tokens between frames — the 11,664 visual tokens occupy an 11,784-wide span broken by 15 gaps, and a patch assuming one contiguous block did nothing. All five would have entered the results tables as working methods under an execution-log check alone.

Measured token budgets

Table 8: Measured visual-token budget on Qwen3-VL-8B, read from each method’s execution log rather than its CLI flag. Input is 11,664 visual tokens at 32 frames. Nominal 15% describes five of ten methods.

Method	Reduction stage	Measured budget
HoliTom	pre-LLM segment retain + in-LLM merge	7.5% (875 tokens)
STTM	quadtree spatio-temporal merge	9.0% (1,050)
FastV	in-LLM attention rerank, layer 2	15.0% (1,750)
FlashVID	pre-LLM temporal-segment merge	15.0% (1,750)
AIM	bipartite merge + PageRank prune	15.0% (1,750)
VisionZip (contextual)	key-similarity merge in vision tower	15.0% (1,750)
DyCoke	temporal merge + KV prune, layer 3	49.0% (5,716)
PruneVID	DPC-KNN cluster merge, layer 10	50.0% (5,832)
MDP3	frame selection (DPP)	8 of 32 frames
VideoITG	grounded frame selection	32 of 512 sampled

Provenance

Every figure in the results tables and all four figures were recomputed for this report directly from each run’s per-sample prediction records, not transcribed from prior documentation. Recomputed accuracy agreed with each run’s stored summary to within 0.01 points across all 42 available

run directories, and independently reproduced the documented divergence counts, paired win/loss counts and per-category figures.

Four LLaVA-OV results — MDP3 (53.06%), AIM (52.86%), VideoITG (52.86%) and STTM (51.72%) — are omitted from Table 3 and all figures because their run directories were absent from the available cache; only non-comparable sampled-decoding variants were present. They appear as dashes in Table 1. The DyTo token-matched control (§2.8) is cited on the same basis.

B Paired-prediction statistics

Differ counts predictions unequal to the backbone’s over all 8,052 rows (accuracy uses the 4,018 scoreable subset); 0 would indicate a silent no-op. *Fixed* and *broke* are the McNemar discordant pairs on scoreable items.

Table 9: Paired-prediction statistics against each backbone.

Method	Differ	Fixed	Broke	χ^2	Significant
<i>LLaVA-OV-7B</i>					
DyCoke	1,031	170	142	2.34	no
FlashVID	1,610	268	242	1.23	no
HoliTom	1,838	299	280	0.56	no
PruneVID-OV	4,536	473	1,054	220.30	yes
FastV	4,605	475	1,113	255.52	yes
<i>Qwen3-VL-8B</i>					
PruneVID	1,074	56	70	1.34	no
DyCoke	1,048	83	110	3.50	no
HoliTom	1,657	131	219	21.63	yes
MDP3	1,626	159	274	30.01	yes
FastV	2,029	149	290	44.65	yes
VisionZip (contextual)	2,035	191	340	41.25	yes
STTM	2,095	170	389	85.02	yes
FlashVID	2,625	178	414	93.29	yes
VideoITG	2,522	206	454	92.44	yes
AIM	2,861	194	457	105.44	yes
<i>Backbone vs. backbone</i>					
Qwen3-VL-8B vs. LLaVA-OV-7B	7,081	942	546	127.47	yes

C Limitations

Scope. All results are on MotionBench and two backbones. Nothing here generalizes to other video benchmarks; the per-category analysis of §2.5 in fact predicts that these methods will look better on benchmarks weighted toward appearance and object recognition.

Single measurement per cell. Greedy decoding makes each configuration exactly reproducible, so there is one measurement per cell and no run-to-run variance estimate. Reported uncertainty is binomial, not empirical — which is why the ± 1.54 -point floor is applied strictly.

Standardization is not paper fidelity. FastV’s published default is $r = 0.5$; the figures here are FastV *at* 10–25%, well outside its validated range, and should not be read as characterizing the

method as published. Since methods without a retention knob keep their defaults, the cross-method comparison is at a common nominal setting and not at equal compute (Appendix A).

Determinism is conditional. Bit-identical reproduction was verified across nodes of the same GPU generation, same software stack, batch size 1. Different GPU generations may perturb near-tied logits; that comparison was not run. Replicators on other hardware should expect close, not identical, numbers.

Zero divergence is strong but not airtight. A method pruning only tokens the model already ignored would execute correctly and produce zero divergence, and the gate would misclassify it. No such case was observed.

Scoring protocol. Accuracy depends on a letter-matching regular expression and on excluding the 4,034 NA items — roughly half the benchmark. Both are documented choices; neither has been ablated.

D Open items

1. **FastV and PruneVID-OV on LLaVA-OV: method or harness?** Both drive reduction through the same code path and fail with matched signatures. The decisive experiment is a `keepall` control that traverses that path while discarding nothing; if it does not return $\approx 52.66\%$, both rows are harness artifacts and must be withdrawn and re-run. Four arms are queued and have not reported. The retention-invariance and answer-distribution evidence of §2.7 stands either way — what is open is whether it describes the published methods or this integration of them.
2. **Sampled-decoding answer distribution.** A companion study under temperature = 0.7, top- p = 0.9 (not comparable to any gated result, since sampling defeats the divergence check) is missing its FlashVID row; that run is in flight.